COMP9434 Report 1

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Problem Statement

The goal of this phase is to deploy and demonstrate a fully functional wall-following and visual mapping system on a real TurtleBot 3 robot.
Specifically, the robot is expected to be capable of:

1. Navigate through physical mazes by relying on reliable wall-following behavior to ensure collision-free movement,
2. Detect and classify cylindrical colored markers with various bicolor combinations,
3. Use Cartographer 2D to accurately record the maze layout and marker positions, and
4. After the cycle is completed, it automatically returns to the initial position.

My primary contribution is to enhance the robustness of color recognition in low-light laboratory environments and optimize motion parameters for smoother navigation.

Background

In the preparation phase, our team reviewed several technical resources to effectively understand and implement the required subsystems.
The TurtleBot3 electronic manual and ROS 2 navigation tutorial provide a deep understanding of the wall tracking algorithm and motion control principles based on LiDAR, which is crucial for wall_follower.cpp nodes **[1] [2]**.
We conducted research on the ROS 2 cartographer documentation to understand map construction, trajectory visualization, and the integration of SLAM with other ROS nodes **[3]**.
The OpenCV HSV color detection tutorial guides the configuration of the hue saturation value threshold for the see_marker.py visual node and explains how lighting changes affect the perceived color range **[4]**.
Finally, we referred to the experimental materials from weeks 3-5 of COMP3431 to understand the implementation structure and testing procedures.
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Broad Explanation of the Team's Solution

Our team has designed an integrated ROS 2 system consisting of three interdependent nodes - wall_follower.cpp, see_marker.py, and point_transformer.py-to - to enable TurtleBot 3 to autonomously navigate through mazes, detect colored cylindrical markers, and work with cartographers to construct a complete occupancy map.
All nodes run simultaneously and communicate through ROS 2 themes to ensure real-time synchronization between perception, motion, and mapping modules.

wall_follower.cpp — Reactive Left-Wall Navigation

The wall_follower node subscribes/scans (LiDAR) and/odm data to maintain a constant distance from the left wall while avoiding frontal collisions.

Each scan is reduced to 12 average distance readings, and the control logic determines the linear velocity and angular velocity based on the wall geometry.

When the current path is blocked, the robot uses a pre adjusted speed threshold to perform evasive turns:

```
if (scan_data_[FRONT] > 0.55) {
   if (scan_data_[LEFT_FRONT] < 0.43 && scan_data_[LEFT_BACK] < 0.43) {
      if (scan_data_[LEFT_FRONT] < 0.35 && scan_data_[FRONT_LEFT] < 0.35) {
        update_cmd_vel(0.1, -0.2);
      }
       update_cmd_vel(0.1,0); //go straight when left back and front are similar
       return;
   }  else if (scan_data_[LEFT_FRONT] > 0.40) {
       update_cmd_vel(0.1, 0.3); //track left wall when too far right
       return;
   } else if (scan_data_[LEFT_FRONT] < 0.40) {
       update_cmd_vel(0.1, -0.3); // turn right when too close left
       return;
   }
   return;
```

This behaviour allowed smooth cornering and robust navigation even in narrow corridors.

During integration, we tuned the **linear speed** and **angular limits** to prevent overshooting turns on the real robot and to minimise jerky motion.

see_marker.py — Marker Detection via HSV Segmentation

The see_marker node processes real-time frames from the TurtleBot's RGB camera (/camera/image_raw), converting them from BGR to HSV space for robust colour segmentation.

Each frame is filtered using in-range thresholds for pink, blue, green, and yellow regions:

```
colours = {
 "pink":   ((140,0,0), (170, 255, 255)),
 "blue":  ((100,0,0), (130, 255, 255)),
 "green": ((40,0,0), (80, 255, 255)),
 "yellow": ((25,0,0), (32, 255, 255))
}
```

For each detected pink spot, the algorithm searches for vertically aligned secondary colors (such as blue, green, or yellow) to form effective marker pairs.

Convert each recognized pair to polar coordinates (distance, angle) using the height and centroid of the blob, and then publish it as a PointStamped message on/marker_position.

I adjusted and expanded these HSV ranges to handle low light indoor conditions, reduce false detections, and improve the stability of markers during motion.

Explain Your Contribution

(a) HSV calibration and labeling robustness

In the simulation, the initial HSV range used for label segmentation is designed under bright and ideal conditions.

However, during laboratory testing, the lighting environment was quite dim, resulting in color distortion, especially for the pink and blue markings, whose tone values shifted towards purple and cyan, respectively.

To address this issue, I conducted several control experiments by capturing sample frames under different lighting intensities and readjusting the HSV boundaries through iterative testing.

I expanded the range of color tones, adjusted the saturation and value thresholds to make segmentation less sensitive to brightness fluctuations.

Those updated thresholds generated clearer binary masks and achieved consistent detection across all six marker combinations (pink on green, green on pink, yellow on pink, pink on yellow, blue on pink, and pink on blue).

I validated the performance by recording the results frame by frame and measuring the detection stability while the robot was moving.

The detection success rate has increased from approximately 70% to over 95%, and the false positive rate can be ignored.

(b) Motion Parameter Optimisation for wall_follower.cpp:

After integrating the perception module, we noticed that the robot occasionally swings around corners or moves too slowly along straight walls.

In order to improve the overall smoothness of navigation, I assisted in adjusting the control parameters in wall_follower.cpp.

Specifically, I:

The linear velocity increases from 0.12 m/s to 0.16 m/s to maintain stable forward motion,

Limit the maximum angular velocity to 1.0 rad/s to prevent excessive turning,

Introduce a speed ramp to limit the acceleration variation between control loops.

(c) Integration Testing and Real-Robot Validation.

Once both subsystems have been tuned, I will help integrate them with the point_transformer node and the Cartographer mapping system.

I attended a real TurtleBot testing conference - launching all nodes through a unified startup file, synchronizing LiDAR, camera, and odometer data, and verifying RViz output.

Through multiple test runs, I have confirmed that the detected markers have been correctly converted to the/map framework and displayed as a MarkerArray overlay that matches the position of the real cylinder.

In these tests, I also monitored the mapping trajectory and ensured that there were no TF lookup delays or mismatched frame IDs, which could distort the global coordinates

of the markers.

How Did You Evaluate Your Method

We evaluated the system through several real-world tests conducted in the maze during week 5. All nodes (wall_follower, see_marker, and point_transformer) run together with the cartographer, and the results are visualized in RViz with grid occupation, trajectory, and marker overlay enabled. This enables us to evaluate navigation smoothness, marker detection accuracy, and map consistency in real-time. After the linear speed and angular speed are adjusted in wall_follower.cpp, the speed of the robot completing the maze is increased by about 12%, and it usually avoids contacting the wall.

The improved HSV range in see_marker.py increases the accuracy of marker detection under dim lighting from approximately 70% to over 90%, providing more stable visual recognition. However, robots still occasionally come into contact with colored cylinders, especially during sharp turns on narrow roads, which may be due to limited side clearance and camera blind spots. Despite this, the overall performance showed **smooth navigation, reliable marker detection, and accurate map projection** during the live demonstration.

Conclusions and Suggestions for Improvements

Overall, the system has achieved most of the project objectives. The robot successfully followed the wall through the entire maze, drew a consistent cartographer's map, and accurately detected colored cylinders. The adjusted HSV range in see_marker.py significantly improves the recognition stability under different lighting conditions, while the motion adjustment in wall_follower.cpp reduces the oscillation and helps maintain a smoother path. The markers are correctly projected into the map frame, and the trajectory displays a continuous closed loop near the starting position, proving that the navigation and drawing modules are well integrated together.

However, there are still some limitations during the live streaming process. Robots still occasionally collide with colored markers, especially during turns or when markers are close to the edge of walls. This may be because the wall tracking logic relies only on the distance of the LiDAR and does not consider the markers as obstacles, coupled with the limited field of view of the camera, sometimes unable to detect nearby cylinders early. Future improvements may include adding an auxiliary short-range sensor (such as ultrasonic or infrared) for real-time obstacle detection, improving angular velocity control with a full PID controller to make turns smoother, and implementing dynamic HSV calibration to automatically adjust thresholds based on current lighting conditions. These enhanced features will further improve reliability, enabling robots to navigate safely even in more complex or uneven lighting environments.

References

[1] Robotis. (2024). *TurtleBot3 e-Manual*. Retrieved from https://emanual.robotis.com/docs/en/platform/turtlebot3
[2] ROS 2 Documentation. (2024). *Navigation Tutorials for TurtleBot3*. Retrieved from https://docs.ros.org/en
[3] Google Cartographer Project. (2024). *Cartographer ROS 2 Documentation*. GitHub Repository: https://github.com/cartographer-project/cartographer_ros
[4] OpenCV Documentation. (2024). *Color Spaces and HSV Thresholding*. Retrieved from https://docs.opencv.org